⛽ Biofuel & Energy: Guided Notes · ANSWER KEY

Biology A · Unit 3 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A3 Matter & Energy, A5 Lipids, A6 Biofuel + Combustion, A7 Energy Reactions, A8 Carbon Cycle, Final Exam

Name: _____

Use the class slides and your reading. Key words:

matter · *energy* · *potential* · *kinetic* · *lipid* · *biodiesel* · *combustion* · *exothermic* · *endothermic* · *renewable* · *carbon cycle*

A. Matter and energy (A3)

1. Matter has **mass** and takes up space.

2. Fill in the table.

Example	Potential or kinetic?
water behind a dam	**potential**
a boulder rolling downhill	**kinetic**
gas in a car's tank	**potential (chemical)**

3. A hot pan warms your hand. Is this a transfer or a transformation? Explain.
Transfer: heat energy moves from the pan to your hand without changing form.

B. Lipids and biofuel (A5, A6)

4. A fat is glycerol plus three fatty **acids** .

5. Lipids store about **2.5** times the energy of sugar.

6. List the energy changes when a bus burns biodiesel.
Chemical energy in the fuel → heat (thermal) → mechanical energy that moves the bus.

C. Reactions and the carbon cycle (A7, A8)

7. A reaction that gets hotter is **exothermic** .

8. Coal, oil and natural gas are **fossil** fuels.

9. Name one process that takes CO2 out of the air and one that puts it back.
Out: photosynthesis. Back: burning (combustion) or breathing (respiration).

🔑 *Key written by our team from Ms. Dunham's materials, please check.*